

Prof. Dr. Lincy Pyl and Selection Committee
Department of Mechanics of Materials and Constructions (MeMC)
Vrije Universiteit Brussel

Subject: Motivation letter – PhD candidate, Mechanics of Materials and Constructions (M²S)
Introduction video: <https://drive.google.com/file/d/14s-V07z3fopEllnUP8vmMJPMxQW57mg9/view>

Dear Prof. Pyl and Selection Committee,

I am Pooja Awasare, a civil and environmental engineer. I am writing to apply for the PhD position in your department at VUB. This letter is my short story of how I arrived here, and why M²S feels like the right place for the next step.

My path started in India with a Diploma and a Bachelor's in Civil Engineering. In those years I spent a lot of time in the materials lab: concrete, steel, soil, aggregates and water. I liked that work more than I expected. When a cube failed, or a soil test did not match what I thought, I did not want to stop at the number. I wanted to understand why. That habit stayed with me. I am a consistent person. When a question opens, I keep digging until I reach the root of the problem. I am curious in the same way: I like to go deep, not only skim the surface.

Later I completed a Master's in Environmental Engineering at HTW Dresden. My thesis looked at how construction teams can manage projects and use resources more carefully. That work pushed me further toward a simple idea: if we want cleaner and more sustainable building practice, we cannot ignore materials and structures. They are the heart of construction. Everything else sits on how materials behave, how we test them, and how we design with them.

That is why this PhD matters to me. I want to spend four focused years on materials and structures at research level, learn advanced testing and modelling, publish carefully, and grow inside a real research group. I have lived and worked in Belgium, including with a construction partner on circular practices in project delivery. I want to build my research life there again, starting in Brussels on 1 October 2026.

I did not choose VUB by name alone. I have been following MeMC and reading about your group's work. I read about Prof. Pyl's long line of research on lightweight structures, on how innovative materials perform under load, and on linking experiments with analytical and numerical models. I also followed MeMC's wider direction on resource-efficient and circular building systems, where structures are made to carry load well while using less material and staying open to reuse. That mix of careful testing, clear mechanics and real construction questions is exactly the kind of depth I want. It matches how I like to work: stay with a problem until the cause is clear.

I am an early-career researcher and ready to apply for fellowships such as FWO. I work calmly, alone and in a team. I welcome teaching support and department life on the Elsene campus. My English is strong (C1), and I will keep improving my Dutch from the A1/A2 level I started in Belgium.

I would be glad to shape a PhD topic with you around the mechanical behaviour of construction materials and more careful structural design for sustainability. Thank you for reading this letter and for watching my short introduction video. I hope we can speak further.

Yours sincerely,

Pooja Awasare